

Searching for the
secrets of the early
Universe



MPhys Public Summary
Clara Pollock 2024

This is our galaxy - The Milky Way.
It is full of stars, planets, and gas, and it is
held together by gravity.

There are billions of galaxies in the
Universe - everywhere we look in the sky!



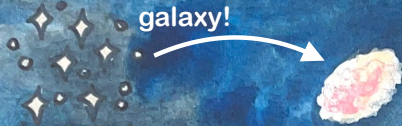
This is where our
Solar System is!



How many
galaxies can you
count here?

There are lots of galaxies now, but something that interests a lot of scientists are the **FIRST** stars and galaxies in the universe, and how they are different from our Milky Way.

Lots of stars
make up a
galaxy!



The Universe is expanding, so these galaxies are very far away. Light travels fast, but it takes 8 minutes to get to Earth from the Sun, and billions of years from these distant galaxies. So we can see galaxies that were born not long after the start of the universe!



8 minutes



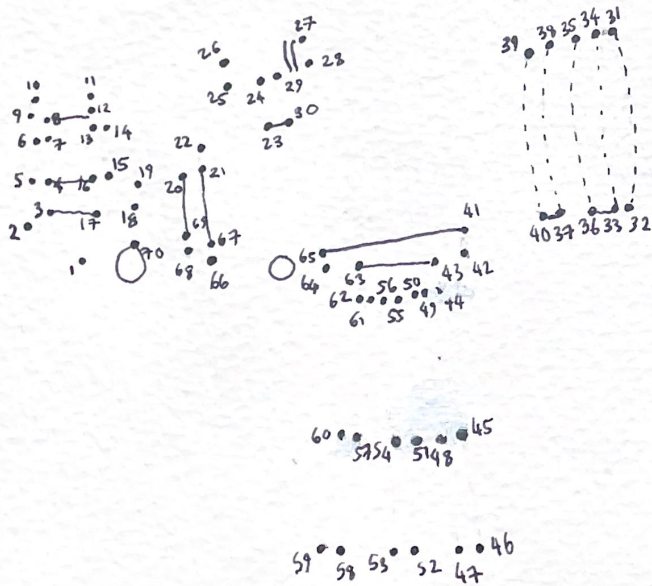
If you have ever looked through a pair of binoculars, you will know that you can see things that are far away much bigger and clearer with them.





The same is true with telescopes!
They let us see stars and galaxies
that are far away!

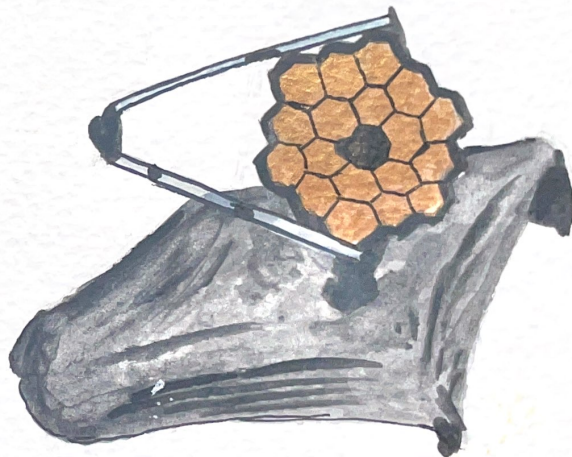
Can you complete the dot-to-dot to
reveal the telescope?



The telescope that is used to look for early
galaxies is the James Webb Space Telescope
(JWST).

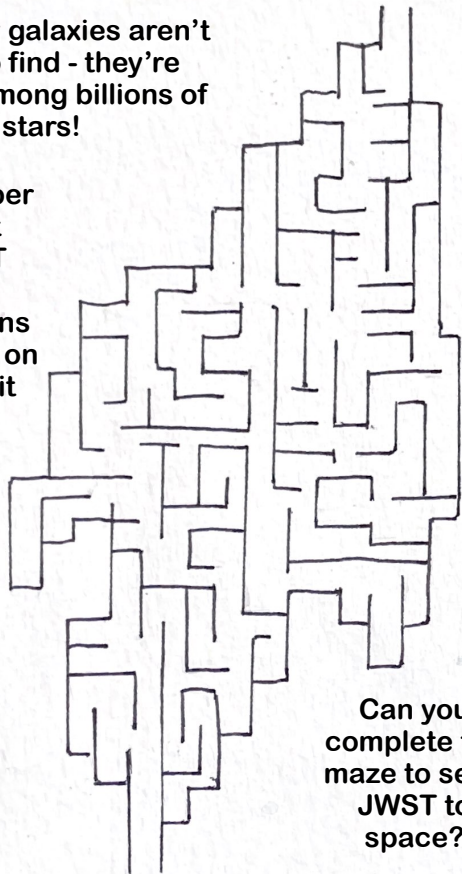
JWST is very special because it can see things
no other telescope can.

JWST can see in infrared, which helps it see
through dust clouds to find the early galaxies!



Our early galaxies aren't easy to find - they're hidden among billions of stars!

It was super important that JWST had no distractions from light on Earth, so it had to be sent to space!



Can you complete the maze to send JWST to space?

Why do we want to find these early galaxies? Well, they might hold clues about how our own galaxy formed, and how it has changed over billions of years.

In the early universe, galaxies were closer together so there were more mergers - galaxies bumping into each other, and maybe forming one big galaxy!

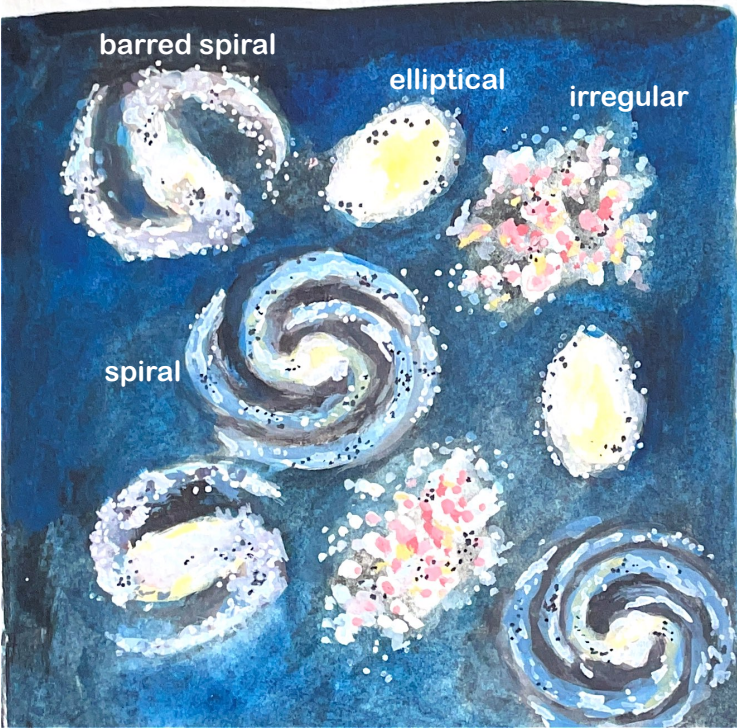
Our galaxy has had several mergers over its life!

We also want to understand how the very first stars formed, and early galaxies can give us clues for this!

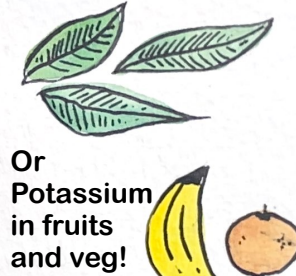


What about shapes? In the early Universe galaxies were still forming, growing, and changing. How did they form into the beautiful shapes we see today?

Can you pair up the galaxy shapes?



Another important thing is the chemical elements. Everything we see is made up of different elements.



Like the Helium that makes balloons float!

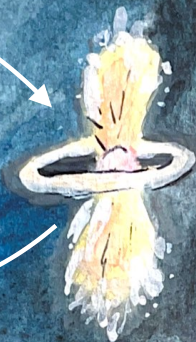


Even the Oxygen in the air that we need to breathe! Oxygen is created in stars.

Oxygen is
made in
stars



When stars reach the end
of their life they explode.



Then it can
be recycled
back into
stars!



Releasing the
elements back
to the galaxy



So if we look for
Oxygen in the early
galaxies, this can
tell us stories about
how the very first
stars were formed!
Perhaps unlocking
a big secret of the
Universe!

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For my niece and nephew
Freya and James